Exploring Decision Boundaries and Calibration through the Penultimate Layer

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Abstract

Recently, there has been significant research on model calibration. However, most studies have focused on which models exhibit better calibration and how to improve calibration, with little attention given to understanding why these phenomena occur. Our research aims to interpret the decision boundaries of classification models based on the penultimate layer, which is the last layer before classification. This is because confidence values are closely related to how close a data is to the decision boundary. We show that decision regions tend to form in the shape of a cone, with apexes centered near the origin. This property is likely to occur due to the neural network weights being initialized randomly near zero, as well as the use of various normalization techniques like batch-norm and layer-norm. Our experiments also show that with cone-shaped decision regions, penultimate layer outputs near the origin hold lower confidence, while outputs that are far from the origin and closer to the center of the cone achieve higher confidence. In addition, our studies suggest that certain regularization methods which are used to aid classification performance can constrain the model's penultimate layer output to be close to the origin. This influences model calibration by making underconfident predictions, as confidence contours of lower confidence is centered near the origin. We believe examining the decision boundaries and data distribution inside them can enhance our understanding of model calibration, explaining why certain models demonstrate better calibration and why regularization can mitigate miscalibration due to overconfident predictions.

1. Introduction

In recent years, deep learning has revolutionized various fields, providing state-of-the-art performance in image classification tasks. However, one critical aspect that remains a challenge is model calibration. Model calibration refers to adjusting a model's predicted probabilities to better align with the actual accuracy of outcomes, ensuring precise confidence levels in its predictions. Having a well calibrated model is important for applications like medical diagnosis and autonomous driving.

Increasing the capacity of classification models has led to a reduction in classification error. However, this improvement often coincides with a decline in model calibration, especially in Convolutional Neural Networks (CNNs) due to their tendency towards overconfident predictions [1]. To alleviate miscalibration as a result of overconfidence, certain regularization techniques, such as label smoothing [2] and mixup [3] have been found to enhance calibration.

Furthermore, recent studies have revealed that newer classification models like Vision Transformers and MLP-Mixers show better calibration compared to CNN based models like ResNet [4].

To address the gap in calibration when using different models and understand why certain regularizations help improve calibration, we examine the representation space of CNNs and Transformers. Intuitively, the confidence of a model on a data that is close to the decision boundary is likely to be lower. Therefore, it is reasonable to empirically study the decision boundary and how data from different classes are distributed inside them to gain insight into model calibration.

We find that decision boundaries are formed in a cone-like structure, with apexes of each cone centered near the origin (see Figure 2). This is likely due to commonly used weight initialization techniques like Xavier, initializing weights to be near zero [5]. Also, various normalization methods like batch-norm and layer-norm also contribute to this property by preventing values from straying too far from zero.

In addition, we show that when models are trained with certain regularization techniques, the distribution of data in the representation space changes, influencing model calibration.

2. Related Work

2.1. Measure of model calibration

Calibration is commonly quantified in terms of Expected Calibration Error (ECE; Naeini et al. 2015), which measures the alignment between a model's predicted probability and actual accuracy [1]. It is defined as

$$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{N} |\mathrm{acc}(B_m) - \mathrm{conf}(B_m)| \,,$$

where N is the number of samples, and M is the number of equally spaced bins partitioning the predictions.

However, because ECE takes the absolute difference between predictive confidence and accuracy, it does not differentiate between miscalibration due to overconfidence and underconfidence. Therefore, signed ECE (sECE) was defined as follows [6].

$$\mathrm{sECE} = \sum_{m=1}^{M} \frac{|B_m|}{N} \big(\mathrm{acc}(B_m) - \mathrm{conf}(B_m)\big)$$

With sECE, miscalibration due to overconfident and underconfident predictions can be distinguished by examining the sign of the value. A positive sECE indicates an underconfident model, while a negative value results from predictions being overconfident.

2.2. Empirical studies on model calibration

In 2017, it was found that modern neural networks are poorly calibrated [1]. They show that as model capacity increases, classification error decreases resulting in better accuracy. However, calibration performance degrades with increased model capacity due to overconfident predictions. Their work also suggests that models trained with less weight decay has a negative impact on calibration. This indicates that weight decay, which is one of the regularization mechanisms, can aid miscalibration due to overconfidence. Other regularization techniques, such as label smoothing [2] and mixup [3] have also been found to improve calibration.

Newer models which achieved state-of-the-art in image classification (e.g., ViT, MLP-Mixers), were shown to be better calibrated despite being designed to focus on accuracy [4]. Their results suggest that model architecture is the major determinant of calibration, rather than model size and pretraining amount.

3. Decision Boundary

In this section, we show that the final layer of classification models, which is usually a linear classifier, exhibits a tendency to divide decision boundaries around the origin, forming cone-like decision regions.

3.1. Experimental Setup

Because the dimension of the output vectors from penultimate layers are very high (e.g., $\mathbf{x} \in \mathbb{R}^{512}$ for ResNet18, and $\mathbf{x} \in \mathbb{R}^{768}$ for ViT-B/16), we create models with an additional linear layer which projects the high dimensional data into a lower 2-dimension for visualization. An overview of our modified model for visualization is shown in Figure 1.

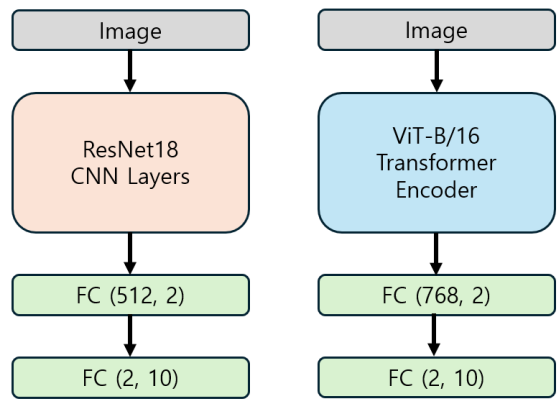

Figure 1. Model overview. The original model for both ResNet18 and ViT-B/16 only has one fully connected (FC) layer, which makes predictions based on the output of CNN layers and Transformer Encoder respectively. However, to visualize the data distribution in the representation space and decision boundaries, we add another linear layer to project the original output of CNN layers and Transformer Encoder into 2-dimensional space before classification.

The two models were trained on the STL-10 dataset, with no augmentation or regularization techniques. For object function optimization, Stochastic Gradient Descent (SGD) was used.

To test the decision boundary properties on the actual models, we use the ResNet50 and ViT-B/16 models trained on the ImageNet dataset. We start by checking the predicted class of linearly interpolated points between the data in the representation space and the origin. The data in the representation space refers to the output vector of the penultimate layer when a single image data is fed into the model, and we will refer to this as the representation vector. For further validation, we repeat the same procedure, but replace the origin with the feature-wise mean of the representation vectors produced by the ImageNet validation data.

3.2. Results

Visualization models. Figure 2 visualizes representation of train data on the 2-d representation space and the decision boundaries. It can be observed that decision boundaries are formed in a cone-like structure, with the apexes of each cone centered near the origin for both ResNet18 and ViT-B/16 models.

Actual models. Figure 3 shows the predicted classes of linearly interpolated points between the representation vectors and the origin/feature-wise mean, when they are fed into the fully connected classification layer. It is evident that for both ResNet50 and ViT-B/16, the ratio of data points predicted to be the same as the representation vectors is much higher when the other end of the

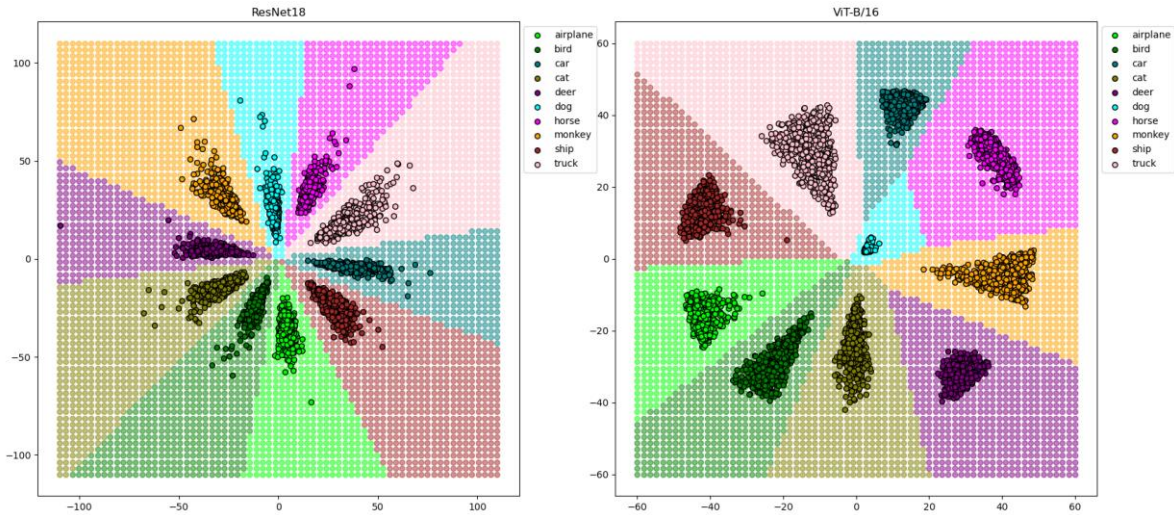

Figure 2. Visualization of STL-10 train data in the representation space and decision boundaries. Each dot with black borders represents a mapping of the input image in the representation space. Dots without borders are arbitrarily created and fed to the final layer to visualize the decision boundaries.

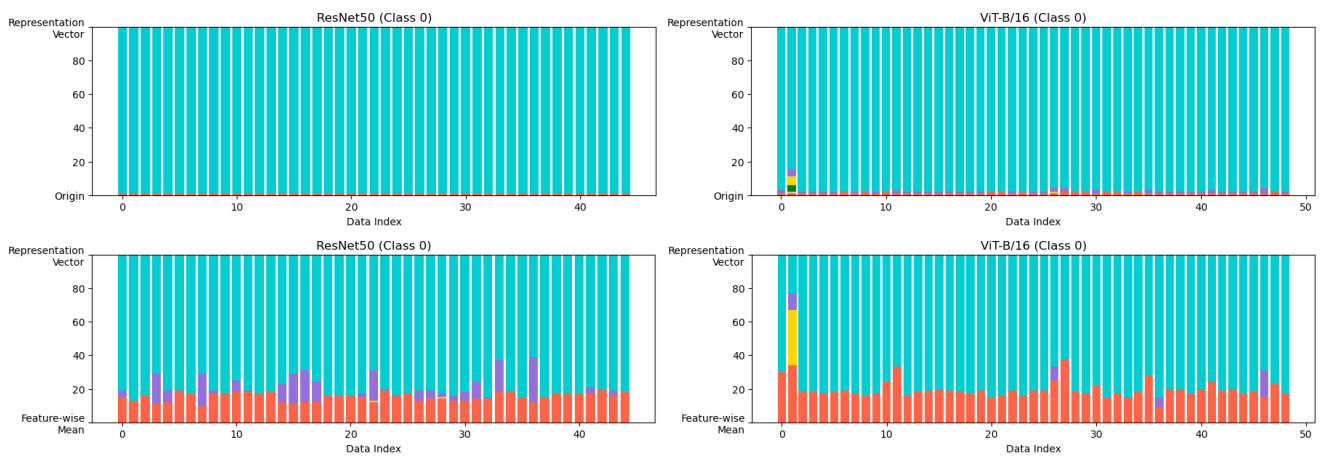

Figure 3. Predictions of the linearly interpolated points between representation vectors and (origin / feature-wise mean). Each bar on the x-axis shows the predicted class of the representation vector, the origin/feature-wise mean, and 100 points created in between. Data points predicted to be the same class as the representation vector (class 0 in this case) are colored in blue, while data points predicted to be the same class as the origin/feature-wise mean are colored in red. Other colors (e.g., purple, yellow) indicate data points predicted to other classes.

interpolation is the origin, compared to the feature-wise mean. Since Figure 3 only plots results from class 0, Figure 4 shows the histogram of the ratio of interpolated data points predicted to be the same as the representation vectors across the validation dataset. We can confirm that for almost all the data, the predictions of interpolated data points between the representation vectors and the origin are the same. This indicates that decision regions which belong to other classes are unlikely to exist in between the origin and the representation vectors. This suggests that classification models learn to transform input images into the correct region in the representation space, where decision boundaries are cone-like structures formed around the origin.

4. Calibration

Despite the significance of model calibration, its analysis from the perspective of decision boundaries has been largely overlooked.

4.1. Confidence values

The confidence values across decision boundaries for Figure 2 is depicted in Figure 5. It is consistent with the intuition that if a data is closer to the decision boundary (edge of the cones), its confidence value is lower.

We explore the confidence values of the actual models trained on the ImageNet dataset with respect to the root mean square (RMS) of the representation vector, and its cosine similarity with the feature-wise mean of the representation vectors predicted to the same class. Referring to the decision boundary properties mentioned in Section 3, we set the starting point of the representation vectors to the origin. Therefore, representation vectors with smaller RMS indicate that they are closer to the origin, and vectors with lower cosine-similarity to the feature-wise mean vector suggest that they are closer to the decision boundary.

Figure 6 shows the results on two different sets of pretrained weights for both ResNet50 and ViT-B/16. V1 weights of ResNet50 are obtained by training without any regularization, except weight decay. However, the V2 weights are acquired by training ResNet50 using label smoothing, mixup, cutmix, etc. For ViT-B/16, the V1 weights are computed based on similar training recipes used for the V2 weights in ResNet50. However, the Swag Linear V1 weights for the Vision Transformer are obtained with less regularization. Compared to weight decay of 0.3 and 0.2 alpha for mixup in V1, it uses smaller values, 1e-4 for weight decay and 0.1 alpha for mixup. In addition, cutmix and label smoothing was not used in Swag Linear V1.

In brief, for ResNet50, V2 is trained with more regularization than V1. For ViT-B/16, V1 is more regularized than Swag Linear V1.

One property shared by all four weight sets is that the cosine similarity between representation vectors and feature-wise class mean vectors positively correlates with confidence values. This indicates that representation vectors which are closer to the center of representation vectors of the same class show higher confidence values.

Comparing results between the two weights of each model, it can be observed that when models are trained with more regularization, the RMS of representation vectors become smaller. Moreover, the overall confidence values decrease as well.

These properties, higher cosine similarity with the class mean vectors showing higher confidence and lower RMS representation vectors showing lower confidence are in line with our theory of cone-shaped decision boundaries.

4.2. Model calibration

The ratio of the amount of data in each confidence bin, and reliability diagrams are also shown in Figure 6. ResNet50 and ViT-B/16 with weights V1 and Swag Linear V1 respectively show overconfidence in their predictions. However, with regularized weights V2 and V1, they become underconfident. This is consistent with the fact that regularization can reduce tendency towards overconfident predictions in classification models.

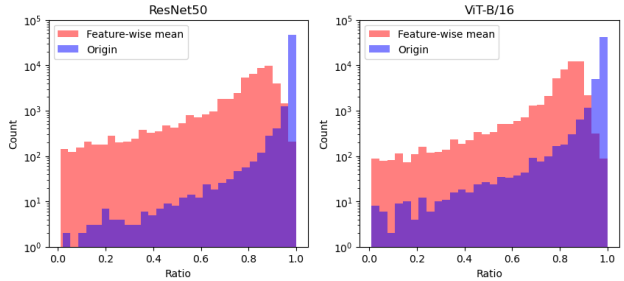

Figure 4. Histogram of the ratio of interpolated data points classified as the same class as the representation vectors.

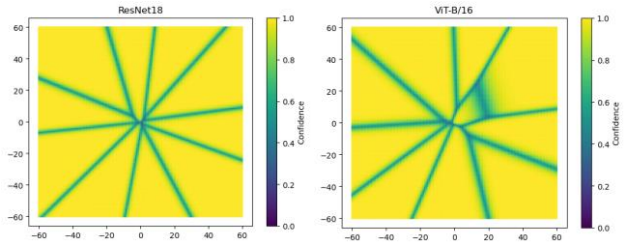

Figure 5. Confidence values across the representation space in Figure 2.

However, although both models showed less overconfidence with regularization, their predictions became too underconfident, that it rather hindered overall calibration. The ECE values increased from 3.67% to 40.74% for ResNet50, and from 1.79% to 5.54% for the Vision Transformer. Therefore, to alleviate miscalibration due to overconfident predictions, regularizations can help, but the degree of regularization should be carefully determined.

Assuming cone-shaped decision boundaries, we can easily test if models become underconfident when the overall RMS of representation vectors decrease by training a linear classifier which classifies different 2-dimensional gaussian distributions. As shown in Figure 7, to imitate the reduction in RMS, we reduce the mean values and standard deviation of the gaussian distribution. A linear classifier trained to classify these distributions makes a cone-shape decision boundary like Figure 5 for both high RMS and low RMS distributions. However, because the decision boundaries are divided around the origin, the confidence values decrease when RMS is lowered, resulting in a highly underconfident model even though the accuracy for both distributions are the same. This explains why predictions can become underconfident when the RMS of representation vectors decrease in Figure 6.

5. Conclusion

We investigated the decision boundaries through the penultimate layer of classification models and examined its relationship with respect to calibration. Our results suggest that regularization can mitigate miscalibration due to overconfident predictions because models trained with

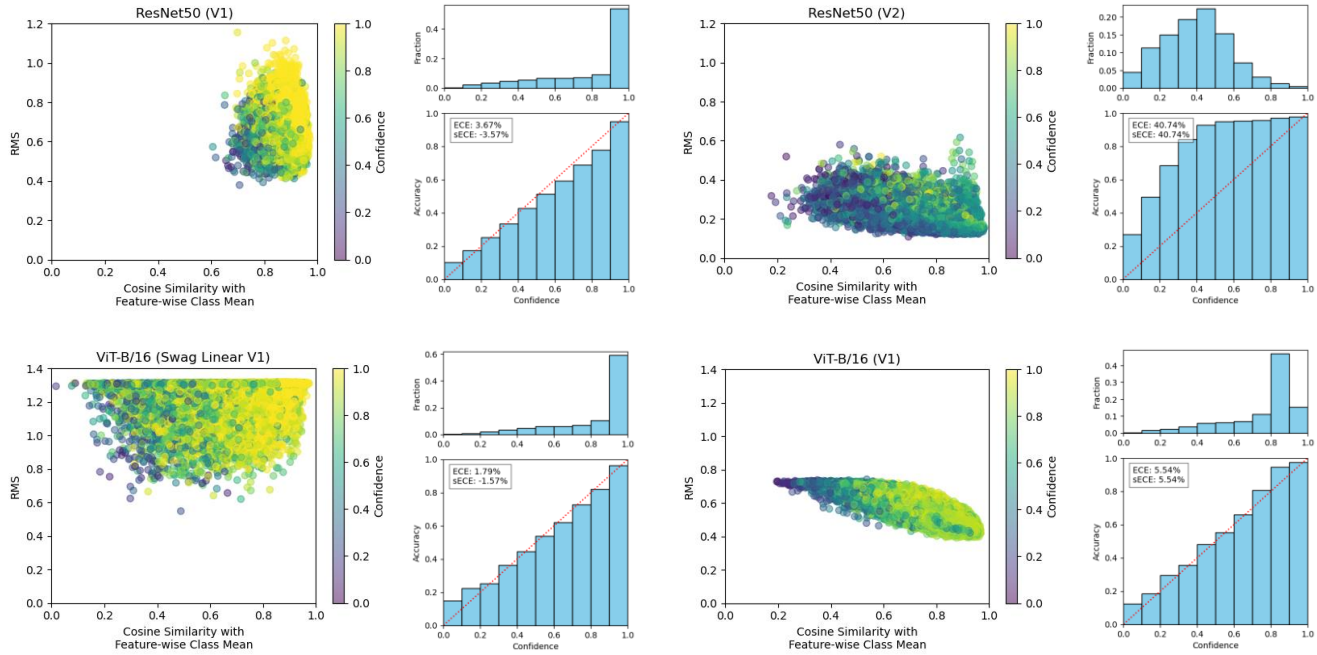

Figure 6. Scatter plots show the confidence values with respect to RMS and cosine similarity between representation vectors and the feature-wise class mean vectors. Each scatter plot has two bar charts on the right. The top chart displays the confidence histogram, while the bottom shows a reliability diagram.

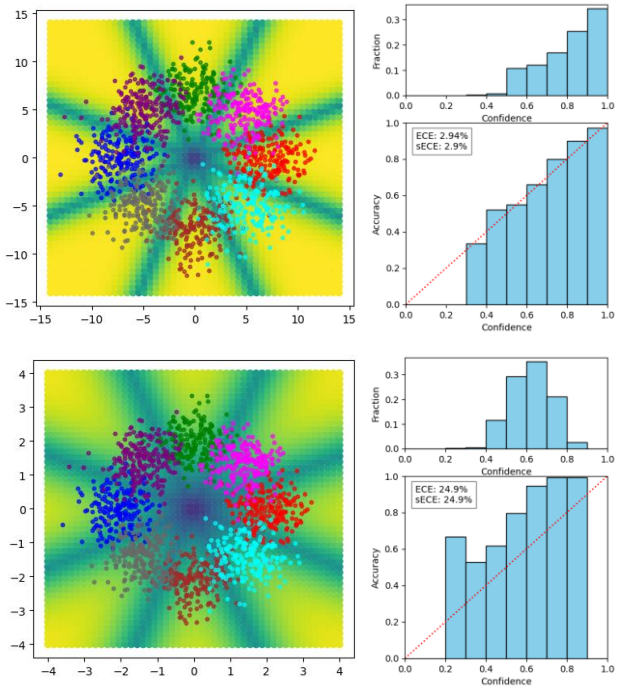

Figure 7. **Left**: Confidence values across the representation space of a linear classifier that classifies different gaussian distributions as scatter plotted. **Right**: confidence histogram and reliability diagram of the linear classifier.

stronger regularization tend to have representation vectors with smaller root mean square (RMS) values. Representation vectors with smaller RMS values are likely to be located closer to decision boundaries, which reduces the likelihood of overconfidence.

6. Future Work

We have discovered that classification models learn to transform input images into the correct region in the representation space, where decision boundaries are cone-like structures formed around the origin. Also, we have shown why regularization can alleviate overconfident predictions. However, it remains future work to understand why boundaries are divided based on the origin, and which regularization in particular is responsible for the decrease in RMS of representation vectors. Additionally, examining the differences in data distribution within the representation space of various models could provide insights into why certain architectures are better calibrated than others.

References

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